

An Investigation into the Semantics of the Dubitative Verb 怀疑(*huaiyi*) in Mandarin Chinese

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A Presentation on JLAO34

July 2021

The Puzzles of Dubitative Predicates in General

- ▶ Debatable precise interpretation
- ▶ Interesting complementation patterns

For example, in English:

- ▶ It seems that *doubt* p is intuitively the antonym of *believe* p , but it actually has a weaker attitudinal commitment than *believe* $\neg p$. (**Anand & Hacquard, 2013**)
- ▶ The verb *doubt* can embed both *that*-clauses and *whether*-clauses, but disallows constituent *wh*-clauses or *whether...or not*-clauses, showing a unique pattern. (**Huddleston, 1994; Karttunen, 1977a, 1977b**)

Another complication in Mandarin Chinese

The dubitative verb in Mandarin Chinese, 怀疑(*huaiyi*), has two empirical interpretations that are nearly opposite:

- ▶ has a tendency not to believe (similar to *doubt* in English, marked as *huaiyi*₁).
- ▶ has a tendency to believe (similar to *suspect* in English, marked as *huaiyi*₂).

An obvious question: Which one above is selected when the verb embeds different types of complements? -The possibility of a disambiguation mechanism.

The main research questions

1. When 怀疑(*huaiyi*) embeds different types of complements, is there a disambiguation mechanism helping people to choose the appropriate one from the two empirical interpretations aforementioned?
2. If there is indeed a disambiguation mechanism, is it possible to combine this mechanism and the two empirical interpretations of the verb into a formal framework, so that a relatively unified semantics of 怀疑(*huaiyi*) can be constructed?

Sentences with non-interrogative complements

- (1) a. 他们 居然 怀疑 这个 项目
they even *huaiyi?* this-CL program
的 可行性。
structural-PTCL feasibility.
'They even *huaiyi?* the feasibility of this program.'
(Yuan, 2014, p. 7)
- b. 人们 怀疑 许多 不明飞行物 与 外星人 有关。
people *huaiyi?* many UFOs with aliens relate.
'People *huaiyi?* that many UFOs are related to aliens.'
(Yuan, 2014, p. 2)
- c. 公司 失窃 了, 我们 很 怀疑
company was stolen -PERF, we strongly *huaiyi?*
是他报 的 案。
be he report structural-PTCL case.
'There was a theft in the company, and we strongly
huaiyi? that it was him who reported the case.'
(J. Zhang & Wu, 2013, p. 2)

Issues of interrogative complements

- ▶ In this dissertation I limit my discussion within two particular types of interrogative complements: those with 'B-not-B' and 'M-not-M' structures.¹
- ▶ Traditionally, they are regarded as variants of a famous question type in Mandarin Chinese, the 'A-not-A questions' or 'reduplicative questions' as named by Zhu (1994, p. 229), involving the structure of 'two copies of a predicate with one copy negated' (Hagstrom, 2006, p. 174).

¹Terminologies adopted from **Hagstrom (2006)**.

Examples of 'A-not-A questions'

- (2) a. 小唐 喜欢 不 喜欢 小强?
Tang like NEG like Qiang?
'Does Tang like Qiang (or not)?'
- b. 小唐 喜 不 喜欢 小强?
Tang li- NEG like Qiang?
'Does Tang like Qiang (or not)?'

The only difference between the two examples above is that (2-a) has reduplicated the whole verb predicate while (2-b) has reduplicated only the first syllable of it, which is not a complete word on its own.

'B-not-B' and 'M-not-M' structures

- ▶ In 'B-not-B' and 'M-not-M' structures, the verb predicate that is copied or reduplicated in 'A-not-A questions' are replaced by a copula and a modal respectively.
- ▶ 'B-not-B' questions are with the structure 是不是 (*shibushi*, be+NEG+be).
- ▶ 'M-not-M' questions can take various forms based on different modals, including 会不会 (*huibuhui*, would/can+NEG+would/can) and 能不能 (*nengbuneng*, can+NEG+can).
- ▶ There are also 'B- 否 (*fou*, NEG)' and 'M-否 (*fou*, NEG)' structures which differ only slightly from 'B-not-B' or 'M-not-M' in the form, preferred in written or formal expressions. It's not problematic to regard them as 'quasi-B-not-B/M-not-M structures' in the semantic sense.

Sentences with 'B-not-B' interrogative complements

- (3) a. 他 怀疑 自己 的 破 表 是
he *huaiyi?* oneself structural-PTCL trashy watch **be**
不 是 已经 停住, 为什么 才 11 点钟 呢?
NEG be already stop, why only 11 o'clock Q?
'He *huaiyi?* whether his trashy watch had stopped, why
was it only 11 o'clock?'

(X. Li, 1987, p. 116)

- b. 这 一切 老式 的 东西
these all old-fashioned structural-PTCL stuff
让 人 怀疑 自己 是否 还 生活 在
make person *huaiyi?* oneself **be NEG** still live in
21 世纪。

21st century.

'All these old-fashioned stuff makes one to *huaiyi?*
whether she still lives in the 21st century.'

(J. Zhang & Wu, 2013, p. 4), simplified.

Sentences with 'M-not-M' interrogative complements

- (4) a. 副团长 对 我
the regimental deputy commander towards I
的 积极性 表示 满意,
structural-PTCL enthusiasm express satisfaction,
但 仍然 怀疑 我 会 不 会 割草。
but still *huaiyi?* I **can NEG can** mow.
'The regimental deputy commander was satisfied with
my enthusiasm, but he still *huaiyi?* whether I can mow.'
(X. Li, 1987, p. 116)
- b. 人们 甚至 开始 怀疑 奥运会 能 否
people even start *huaiyi?* the Olympics **can NEG**
按时 召开。
on time happen.
'People even started to *huaiyi?* whether the Olympics
can happen on time.'
(Yuan, 2014, p. 4), simplified.

Literature review: 怀疑+non-interrogative complements

Syntactic status of the complement? X. Li (1987)

When the complement is nominal or pronominal, 怀疑(*huaiyi*) is interpreted as *huaiyi*₁.

- (5) a. 他们居然怀疑这个项目的可行性。
they even *huaiyi*₁ this-CL program
structural-PTCL feasibility.
'They even doubt the feasibility of this program.'
- b. 如果有谁怀疑地心说,
if have someone *huaiyi*₁ the geocentric theory,
那就是亵渎神灵。
then would be profane god.
'If someone doubts the geocentric theory, then it would be regarded as a profanation of the god.'

Literature review: 怀疑+non-interrogative complements

Syntactic status of the complement? X. Li (1987)

When the complement is clausal, 怀疑(*huaiyi*) is interpreted as *huaiyi*₂.

- (6) a. 我 怀疑 他 去了 医院。
I *huaiyi*₂ he go-PERF hospital.
'I suspect that he has gone to the hospital.'
(Feng, 2017, p. 181)
- b. 人们 怀疑 许多 不明飞行物 与 外星人 有关。
people *huaiyi*₂ many UFOs with aliens relate.
'People suspect that many UFOs are related to aliens.'

Literature review: 怀疑+non-interrogative complements

Controversies regarding the clausal complement:

Han (2001): when the focus is in the clausal complement (like (7-a) and (7-b)), *huaiyi* is interpreted as *huaiyi*₂; while when the focus is in the verb *huaiyi* itself (like (8)), it is interpreted as *huaiyi*₁.

- (7) a. 我 怀疑 明天 会 晴。
I *huaiyi*₂ tomorrow will sunny.
'I suspect that it will be sunny tomorrow.'
- b. 我 怀疑 明天 会 晴。
I *huaiyi*₂ tomorrow will sunny.
'I suspect that it will be sunny tomorrow.'
- (8) 我 怀疑 明天 会 晴。
I *huaiyi*₁ tomorrow will sunny.
'I doubt that it will be sunny tomorrow.'

Literature review: 怀疑+non-interrogative complements

Controversies regarding the clausal complement:

Han (2001), Feng (2017): if 怀疑(*huaiyi*) is modified by adverbs of degree, it's more likely to be interpreted as *huaiyi*₁?

- (9) a. 我 很 怀疑 她 不是 谢大侠
I strongly *huaiyi*₂ she NEG-be Hero Xie
的 女儿.
structural-PTCL daughter.
'I strongly suspect that she is not Hero Xie's daughter.'
- b. 我 很 怀疑 一个 整天 想着 当
I strongly *huaiyi*₁ a-CL all day dream about being
元帅 的 士兵 会 真的
marshal structural-PTCL soldier would really
在战斗中 奋不顾身.
in battles fight bravely.
'I strongly doubt that a soldier who dreams about being a marshal all day would fight bravely in battles.'

Literature review: 怀疑+non-interrogative complements

Controversies regarding the clausal complement:

J. Zhang and Wu (2013): the interpretation is related to different contexts, or more specifically, the 'belonging relationship' between the proposition in the complement and the agent of 怀疑.

- ▶ *huaiyi*₁: Subject(Agent)+*huaiyi*₁+Complement (containing a proposition made by others or the agent herself in the past)
- ▶ *huaiyi*₂: Subject(Agent)+*huaiyi*₂+Complement (containing a proposition made by the agent herself at the point of utterance)

- (10) 公司 失窃 了, 我们 很
company was stolen -PERF, we strongly
怀疑 是他报 的 案.
*huaiyi*₁/*huaiyi*₂ be he report structural-PTCL case.
'There was a theft in the company, and we strongly
doubt/suspect that it was him who reported the case.'

Literature review: 怀疑+non-interrogative complements

Controversies regarding the clausal complement:

Yuan (2014): the emotive evaluation regarding the proposition from the complement distinguishes the two interpretations.

- ▶ When the complement is evaluated as positive or neutral, *huaiyi*₁ is chosen.
- ▶ When the complement is evaluated as negative, *huaiyi*₂ is chosen.

Literature review: 怀疑+non-interrogative complements

- (11) a. 曾 有 人 怀疑 汉语
in the past have people *huaiyi*₂ Chinese language
无 词类 划分。
don't have parts of speech division.
'There used to be people who suspect that there are no divisions of parts of speech in the Chinese language.'
- b. 曾 有 人 怀疑 汉语
in the past have people *huaiyi*₁ Chinese language
有 词类 划分。
have parts of speech division.
'There used to be people who doubt that there are divisions of parts of speech in the Chinese language.'
(Yuan, 2014, p. 10)

Literature review: 怀疑+non-interrogative complements

Counterexamples to Yuan (2014)'s idea:

- (12) a. 我怀疑 他说的 是实话。
I *huaiyi*₂ he say structural-PTCL be truth.
'I suspect that what he said is the truth.'
- b. 我怀疑 他去了 市场。
I *huaiyi*₂ he go ASP market.
'I suspect that he has gone to the market.'

In (12-a), the complement embedded by *huaiyi* can be regarded as positively-evaluated, but it's perfectly possible to interpret *huaiyi* in this sentence as *huaiyi*₂. Similarly in (12-b), the complement can be regarded as neutrally-evaluated, and *huaiyi* in this sentence can also be interpreted as *huaiyi*₂.

Literature review: 怀疑+non-interrogative complements

Controversies regarding the clausal complement:

Lu (2016): the key to the interpretation lies in the relationship between the explicit content proposition in the complement and the immediate contextual evidence (also a proposition, labelled as P, can be implicit and non-verbal).

(13) a. 我 怀疑 明天 会 晴。

I *huaiyi*₁ tomorrow will sunny.

'I doubt that it will be sunny tomorrow.'

the immediate contextual evidence P: the weather in the evening is gloomy.

b. 我 怀疑 明天 会 晴。

I *huaiyi*₂ tomorrow will sunny.

'I suspect that it will be sunny tomorrow.'

the immediate contextual evidence P: there is sunset glow in the evening.

Literature review: 怀疑+non-interrogative complements

If there is a reliable pre-existing logical reasoning ' $P \rightarrow \text{might } Q$ ':

The weather in the evening is gloomy. (P_1) $\rightarrow$ It might be rainy tomorrow. (might Q_1)

There is sunset glow in the evening. (P_2) $\rightarrow$ It might be sunny tomorrow. (might Q_2)

- ▶ In (13-a), the proposition in the complement is not consistent with Q_1 , *huaiyi*₁ is revealed.
- ▶ In (13-b), the proposition in the complement is consistent with Q_2 , *huaiyi*₂ is revealed.

Literature review: 怀疑+non-interrogative complements

A hierarchy and combination of interpreting strategy:

- ▶ From the perspective of valid coverage and disambiguation force, the ranking is logical relationship > pragmatic evaluation > syntactic status.
- ▶ From the perspective of operability, the ranking is syntactic status > pragmatic evaluation > logical relationship.
- ▶ 3 strategies are combined in the process of the disambiguation and interpretation of the verb.

‘怀疑+non-interrogative complements’: a new proposal?

Many clues observed and comments given in the aforementioned literature can lead to a common theoretical framework: the information structure. Two important notions can be relevant here – **informational focus** and **givenness**.

- ▶ Informational focus: a part of the sentence that carries new or prominent information, can correspond to the answer to a wh-constituent constituent question (Halliday, 1967). Expressed by phonetic stress/enlarged pitch range/longer duration while leaving the element in-situ in Mandarin Chinese (Badan, 2015; K. Li, 2009; H. Zhang & Pan, 2019).
- ▶ Givenness: a property indicating that the denotation of a constituent has either been introduced explicitly or implicitly into the discourse. This property can be revealed in several means (de-accentuation/pronominalization/definite references)(Hinterwimmer, 2012; Schwarzschild, 1999).

Focus semantic value: a way to formalize informational focus

Rooth (1992, 2016, 1985):

- ▶ The focus marking introduces a contextually-restricted set of alternatives to the focused constituent, with the original denotation of this constituent (the ordinary semantic value) included.
- ▶ A recursive strategy of 'alternatives all the way down' in which non-focused phrases are with the FSV of a singleton set containing their OSV.

- (14)
- $\llbracket \text{married Bertha} \rrbracket^O = \lambda x. \lambda w. \text{married}(x, \text{Bertha})(w)$
 - $\llbracket [_F \text{married Bertha}] \rrbracket^F = \{ \lambda x. \lambda w. \text{married}(x, \text{Bertha})(w), \lambda x. \lambda w. \text{won}(x, \text{Nobel Prize})(w), \lambda x. \lambda w. \text{sang}(x)(w), \dots \}$
 - $\llbracket \text{married Bertha} \rrbracket^F = \{ \lambda x. \lambda w. \text{married}(x, \text{Bertha})(w) \}$

adopted from Hinterwimmer (2012, p. 1878), with alternations.

Revisiting examples

(15) 我怀疑 [_F明天 会 晴]。

I *huaiyi*₂ tomorrow will sunny.

'I suspect that it will be sunny tomorrow.'

The 'givenness' status of the proposition: not given

$\llbracket \text{怀疑 } [\text{明天会晴}] \rrbracket^F = \{\lambda w \lambda x. x \text{ } huaiyi \text{ that it will be sunny tomorrow in } w, \lambda w \lambda x. x \text{ } huaiyi \text{ that Hong can't drink in } w, \lambda w \lambda x. x \text{ } huaiyi \text{ that Ming has left in } w, \dots\}$

$\llbracket [\text{明天会晴}] \rrbracket^F = \{\lambda w. \text{ It will be sunny tomorrow in } w, \lambda w. \text{ Hong can't drink in } w, \lambda w. \text{ Ming has left in } w, \dots\}$

$\Rightarrow | \llbracket [\text{明天会晴}] \rrbracket^F | > 1$

The chosen interpretation: *huaiyi*₂

Revisiting examples

(16) 我 [_F怀疑] 明天 会 晴。

‘I doubt (*huaiyi*₁) that it will be sunny tomorrow.’

The ‘givenness’ status of the proposition: given

$\llbracket [\text{怀疑}] \text{ 明天会晴} \rrbracket^F = \{ \lambda w \lambda x. x \text{ } huaiyi \text{ that it will be sunny tomorrow in } w, \lambda w \lambda x. x \text{ ‘believes’ that it will be sunny tomorrow in } w, \lambda w \lambda x. x \text{ ‘predicts’ that it will be sunny tomorrow in } w, \dots \}$

$\llbracket \text{明天会晴} \rrbracket^F = \{ \lambda w. \text{ It will be sunny tomorrow in } w \}$

$\Rightarrow | \llbracket \text{明天会晴} \rrbracket^F | = 1$

The chosen interpretation: *huaiyi*₁

The mechanism disambiguating the two interpretations of *huaiyi* in ‘*huaiyi*+non-interrogative complement’ cases lies in the number of members in the set denoted by the proposition’s (equivalent to CP in this case) focus semantic value: when $| \llbracket \text{CP} \rrbracket^F | > 1$, *huaiyi*₂ is chosen; while when $| \llbracket \text{CP} \rrbracket^F | = 1$, *huaiyi*₁ is chosen.

Literature review: 怀疑(huaiyi)+B-not-B/M-not-M complement

Before getting to the discussion about the disambiguation mechanism, there is yet a **compositional problem** in the first place – **what exactly is combined with 怀疑(huaiyi)?**

- ▶ Hamblin (1958, 1973): interrogative complements → questions → sets of propositions.
- ▶ Hagstrom (2006, 2003): A question with an 'A-not-A', 'B-not-B' or 'M-not-M' structure can be regarded as a proposition set containing one proposition and its negation.

Is it the whole set or one of the propositions in the set that is combined with 怀疑(huaiyi)?

Literature review: 怀疑(*huaiyi*)+B-not-B/M-not-M complement

Some implicit claims about the compositional problem can be observed.

- ▶ J. Zhang and Wu (2013): '*huaiyi*₁ (doubt)-**wonder**-*huaiyi*₂ (suspect)' continuum – a neutral uncertainty.

(17) 这 一切 老式 的 东西 让
these all old-fashioned structural-PTCL stuff make
人 怀疑 自己 是否 还 生活 在 21
person **wonder?** oneself **be NEG** still live in 21st
世纪。
century.

Essentially this is claiming that 怀疑(*huaiyi*) is combined with the whole proposition set.

Literature review: 怀疑(*huaiyi*)+B-not-B/M-not-M complement

- Yuan (2014): The disambiguation mechanism is parallel to what is claimed in ‘怀疑+non-interrogative complements’ cases – **when p is positively or neutrally evaluated, $huaiyi_1 + p / huaiyi_2 + \neg p$; when p is negatively evaluated, $huaiyi_2 + p / huaiyi_1 + \neg p$.**

- (18) 人们 甚至 开始 怀疑 奥运会 能 否
people even start *huaiyi*? the Olympics **can NEG**
按时 召开。
on time happen.
‘People even start to suspect that the Olympics can’t happen on time.’

Literature review: 怀疑(*huaiyi*)+B-not-B/M-not-M complement

- X. Li (1987): 'B-not-B' or 'M-not-M' structures are essentially euphemisms to convey some 'undesirable' declarative message with the affirmative proposition (p) or its negation within the sets they denote.

(19) 他 怀疑 自己 的 破 表 是
he *huaiyi*? oneself structural-PTCL trashy watch **be**
不 是 已经 停住, 为什么 才 11 点钟 呢?
NEG be already stop, why only 11 o'clock Q?
'He suspected that his trashy watch had already stopped,
why was it only 11 o'clock?'

Literature review: 怀疑(*huaiyi*)+B-not-B/M-not-M complement

- (20) a. 小黄 怀疑 小唐 是 不 是 喜欢 小强。
Huang *huaiyi*? Tang be NEG be like Qiang.
- b. 小唐 怀疑 小邱 会 不 会 弹
Tang *huaiyi*? Qiu can/would NEG can/would play
吉他。
the guitar.

Two examples above can have both the '*huaiyi*₁ + *p*/*huaiyi*₂ + $\neg p$ ' and '*huaiyi*₂ + *p*/*huaiyi*₁ + $\neg p$ ' interpretations.

Focusing on the compositional problem

- ▶ There is an intuitive similarity between 'B-not-B/M-not-M' complement in Mandarin Chinese and *whether ... or whether ... or not* structure in English.
- ▶ There are many insights about *doubt whether* structure in English that can be useful for our problem.
- ▶ It would be helpful to find out 'B-not-B/M-not-M' complement is more similar to which one among *whether ... or whether ... or not* structures in the syntactic and semantic nature.

Interesting grammaticality patterns in English

- (21)
- a. Ann doubts that it is raining.
 - b. Ann doubts whether it is raining.
 - c. *Ann doubts whether it is raining or not.
 - d. *Ann doubts what the weather is.
- (22)
- a. Ann knows that it is raining.
 - b. Ann knows whether it is raining.
 - c. Ann knows whether it is raining or not.
 - d. Ann knows what the weather is.

Interesting grammaticality patterns in English

Karttunen (1977a, 1977b):

- ▶ Unlike most verb predicates, *doubt* doesn't possess the conformity of the grammaticality when embedding *wh*-questions and *whether*-questions (as 'yes/no' questions, a type of 'degenerated' alternative questions). It can embed *whether*-questions but not *wh*-questions.
- ▶ Observed that disjunctive *whether*-questions (*whether...or...*) and 'extended' *whether*-questions (*whether...or not*) cannot be embedded by *doubt*. Therefore claiming that when embedded by *doubt*, *whether*-complements are not taken as genuine questions and *doubt* is regarded to be outside the category of 'question-embedding verbs'.

Interesting grammaticality patterns in English

Biezma (2009) and Biezma and Rawlins (2012):

- ▶ It's claimed that *whether*-questions and 'extended' *whether*-questions essentially belong to two different categories. The former is analyzed as embedded polar questions ($P_{OL}Qs$), while the latter is regarded as a special type of embedded alternative questions formed by two opposite alternatives ($A_{LT}Q_VNs$), with different question operators.
- ▶ $P_{OL}Qs$ (including *whether* ... structures) are specially treated as singleton sets, mainly out of two reasons: (i) they compositionally lack any alternative-introducing item, for instance disjunctions or other devices of disassociation; (ii) the higher utility value of (or the speaker's preference for) the content proposition than its negation (van Rooy & Safárová, 2003).

Interesting grammaticality patterns in English

2 different question operators proposed by Biezma and Rawlins (2012):

(23) a. **Polar Question Operator:**

$$\llbracket \llbracket [Q] \alpha \rrbracket \rrbracket^c = \llbracket \alpha \rrbracket^c$$

defined only if

(i) $\llbracket \alpha \rrbracket^c \subseteq \text{SalientAlts}(c)$ or if $\text{SalientAlts}(c) = \emptyset$, and

(ii) $|\llbracket \alpha \rrbracket^c \cup \text{SalientAlts}(c)| > 1$

b. **Closure Operator for Alternative Questions:**

$$\llbracket \llbracket [Q] \alpha \rrbracket_{H*L-L} \rrbracket^c = \llbracket \llbracket [Q] \alpha \rrbracket \rrbracket^c$$

defined only if $\text{SalientAlts}(c) = \llbracket \llbracket [Q] \alpha \rrbracket \rrbracket^c$

(Presupposition: listed alternatives have exhausted all salient alternatives)

Constraint: α must contain a disjunction.

Interesting grammaticality patterns in English

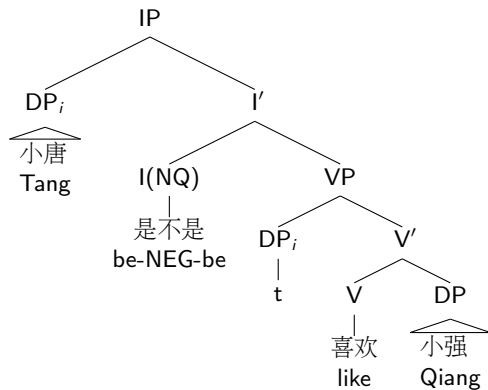
An alternative explanation given by Pruitt and Roelofsen (2011):

All verbs are question-embedding and $P_{OL}Q$ s are still sets with two members, but there is a 'highlighting mechanism' depending on their surface form. Coincidentally, *doubt* is a verb that semantically requires to be combined with the one and only highlighted proposition from the set denoted by the embedding interrogative complement.

- (24)
- a. John doubts whether the door is open.
 - b. John doubts whether the door is closed.
 - c. *John doubts whether the door is open or closed.
 - d. *John doubts whether the door is open or not.
- (25)
- a. John knows whether the door is open.
 - b. John knows whether the door is closed.
 - c. John knows whether the door is open or closed.
 - d. John knows whether the door is open or not.

The similarity between 'B-not-B/M-not-M' complement and *whether* ... structure

(26)



[[是不是_Q[小唐_t 喜欢 小强]]

The similarity between 'B-not-B/M-not-M' complement and *whether* ... structure

- ▶ From the dimension of logical form, 'B-not-B/M-not-M' can be analyzed as scoping over a proposition as a question operator, just as *whether* does.
- ▶ McCawley (1994) claims that the negative morpheme in its reduplicative form is real in the morphological sense.
- ▶ Hagstrom (2006, p. 179) claims that it 'does not decisively present in syntax and interpretation' – not showing the 'cornering effect' proposed by Biezma and Rawlins (2012) as a differentiating trait between $A_{LT}Q_VNs$ from $P_{OL}Qs$.

The rationale of highlighting mechanism in Pruitt and Roelofsen (2011) can be adopted.

The disambiguation mechanism of 怀疑(huaiyi)+B-not-B/M-not-M complement

- (27) 小黄 怀疑 小唐 是 不 是 [_F喜欢] 小强.
Huang *huaiyi*₂ Tang be NEG be like Qiang.
'Huang suspects that Tang likes Qiang.'
The 'givenness' status of the proposition: not given
[[是不是_Q [小唐 [_F喜欢] 小强]]^F
={λw. Tang likes Qiang in w, λw. Tang doesn't likes Qiang in w},
{λw. Tang hates Qiang in w, λw. Tang doesn't hate Qiang in w}, ...}
⇒ | [[是不是_Q [小唐 [_F喜欢] 小强]]^F | > 1
The chosen interpretation: *huaiyi*₂

The disambiguation mechanism of 怀疑(huaiyi)+B-not-B/M-not-M complement

(28) 小黄 怀疑 小唐 [_F是 不 是] 喜欢 小强.

‘Huang doubts (*huaiyi*₁) that Tang likes Qiang.’

The ‘givenness’ status of the proposition: given

$\llbracket [\text{是 不是}_Q] [\text{小唐 喜欢 小强}] \rrbracket^F$

$= \{ \{ \lambda w. \text{Tang likes Qiang in } w, \lambda w. \text{Tang doesn't like Qiang in } w \} \}$

$\Rightarrow | \llbracket [\text{是 不是}_Q] [\text{小唐 喜欢 小强}] \rrbracket^F | = 1$

The chosen interpretation: *huaiyi*₁

The criterion regarding the value of $| \llbracket \text{CP} \rrbracket^F |$ discussed in 怀疑(*huaiyi*)+non-interrogative complement seems to be still working.

The semantics of *doubt* in English

Anand and Hacquard (2013, p. 36), simplified:

- (29) $\llbracket a \text{ doubts}_C \text{ that } \varphi \rrbracket^{c,w,S,g}$ is defined iff
 φ -verifiers in $S' \neq \emptyset \wedge \varphi$ -falsifiers in $S' \neq \emptyset$. **[Uncertainty Condition]**
If defined = 1 iff
 $\exists w' \in S' : [\llbracket \varphi \rrbracket^{c,w',S,g} = 1] \wedge$ **[Doxastic Assertion]**
 p -falsifiers $>_{PROB_{a,w}}$ p -verifiers **[Preference Assertion]**

A proposal for the semantics of 怀疑(huaiyi)

For the verb 怀疑(huaiyi) in the Mandarin natural language, define its metalanguage definition as BIASED-BELIEF.

► **The definition in the metalanguage:**

BIASED-BELIEF(A, p, w) is defined iff

p -verifiers in $DOX_{A,w} \neq \emptyset \wedge p$ -falsifiers in $DOX_{A,w} \neq \emptyset$.

[Uncertainty Condition]

If defined = 1 iff

$\exists w' \in DOX_{A,w} : [p(w') = 1] \wedge$

p -falsifiers $\neq_{PROB_{A,w}}$ p -verifiers

[Doxastic Assertion]

[Preference Assertion]

A proposal for the semantics of 怀疑(*huaiyi*)

- ▶ **The semantics of *huaiyi*:**

$\llbracket A \text{ } huaiyi \text{ } \varphi \rrbracket^w = \text{BIASED-BELIEF } (A, \iota p[\text{HIGHLIGHT}(p) \wedge p \in \llbracket \varphi \rrbracket], w)$

- ▶ **The disambiguation mechanism:**

a. When φ is a clausal complement and $|\llbracket \varphi \rrbracket^F| = 1$, $\llbracket huaiyi \rrbracket = \llbracket huaiyi_1 \rrbracket$ i.e. $\exists w' \in DOX_{A,w} : [p(w) = 1] \wedge p\text{-falsifiers} >_{PROB_{a,w}} p\text{-verifiers}$, where $p = \iota p[\text{HIGHLIGHT}(p) \wedge p \in \llbracket \varphi \rrbracket]$.

b. When φ is a clausal complement and $|\llbracket \varphi \rrbracket^F| > 1$, $\llbracket huaiyi \rrbracket = \llbracket huaiyi_2 \rrbracket$ i.e. $\exists w' \in DOX_{A,w} : [p(w) = 1] \wedge p\text{-verifiers} >_{PROB_{a,w}} p\text{-falsifiers}$, where $p = \iota p[\text{HIGHLIGHT}(p) \wedge p \in \llbracket \varphi \rrbracket]$.

A link with Mari(2016)

- ▶ **Expressive-‘believe’**: a solipsistic mental state, purely conveying a doxastic certainty.
- ▶ **Inquisitive-‘believe’**: The mixture of a doxastic certainty and an epistemic uncertainty.²
- ▶ **‘doubt’**: The uncertainty on both the doxastic and epistemic layers?

²Mari, A. (2016). Assertability conditions of epistemic (and fictional) attitudes and mood variation. *Proceedings of SALT 26*, 61-81.

A possible account

- (30) 他 怀疑 自己 的 破 表 是
he *huaiyi*₂ oneself structural-PTCL trashy watch be
不 是 已经 [_F停住], 为什么 才 11 点钟 呢?
NEG be already stop, why only 11 o'clock Q?
'He suspected that his trashy watch has stopped, or else
why was it only 11 o'clock?'

The 'givenness' status of the proposition: not given

[[是不是_Q [破表 已经 [_F停住]]]^F

= { { λw . The trashy watch has stopped in w , λw . The trashy watch has not stopped in w }, { λw . The trashy watch was broken in w , λw . The trashy watch was not broken in w }, ... }

$\Rightarrow | \text{[[是不是}_Q \text{ [破表 已经 } [\sub{F}\text{停住}]]] }^F | > 1$

The chosen interpretation: *huaiyi*₂

A possible account

- (31) 他 怀疑 自己 的 破表 [_F是 不
‘He doubted (*huaiyi*₁) that his trashy watch has
是] 已经 停住。
stopped.’

The ‘givenness’ status of the proposition: given

$[[[_F\text{是不是}_Q] [\text{破表 已经 停住}]]^F$

$= \{ \{ \lambda w. \text{The trashy watch has stopped in } w, \lambda w. \text{The trashy watch has not stopped in } w \} \}$

$\Rightarrow | [[[_F\text{是不是}_Q] [\text{破表 已经 停住}]]^F | = 1$

The chosen interpretation: *huaiyi*₁

A possible account

- (32) a. 我 [_F怀疑] 明天 会 晴。
I *huaiyi*₁ tomorrow will sunny.
'I doubt that it will be sunny tomorrow.'
the implicit immediate contextual evidence P:
the weather in the evening is gloomy.
- b. 我 怀疑 [_F明天 会 晴]。
I *huaiyi*₂ tomorrow will sunny.
'I suspect that it will be sunny tomorrow.'
the implicit immediate contextual evidence P:
there is sunset glow in the evening.

A brief summary of conclusions

- ▶ 怀疑(*huaiyi*), as a dubitative verb in Mandarin Chinese, also possesses a hybrid semantics. Specifically, it existentially quantifies over one's doxastic state, allowing some worlds to falsify the content proposition. But there is also a preference assertion regarding the probability, conveying that there is a bias for the agent such that the probability of being compatible with the real world regarding verifying worlds and falsifying worlds are not equal. The direction towards which this bias leans is determined by the disambiguation mechanism.
- ▶ There is a widely-applicable disambiguation mechanism to distinguish the two interpretations of *huaiyi*₁ and *huaiyi*₂ based on the information structure, mainly depending on the location of the informational focus and the 'givenness' status of the content proposition in the embedding complement.

A brief summary of conclusions

- ▶ On the formal dimension, this mechanism is revealed in the value of $|\llbracket \varphi \rrbracket^F|$. When the informational focus is outside the content proposition and the proposition is marked as 'given', $|\llbracket \varphi \rrbracket^F| = 1$ and $huaiyi_1$ is chosen for the interpretation; when the informational focus is within the content proposition and the proposition is marked as 'not given', $|\llbracket \varphi \rrbracket^F| > 1$ and $huaiyi_2$ is chosen for the interpretation.
- ▶ This mechanism is derived from the logical relationship between the immediate contextual evidence and the content proposition: the former is the case when the proposition is not consistent with the reasoning from the immediate contextual evidence, while the latter is the case when the proposition is consistent with the reasoning from the immediate contextual evidence.

Remaining issues

- ▶ In the proposed disambiguation mechanism for 怀疑(*huaiyi*), nominal complements are not included. The reason is that when embedding a nominal complement, the verb seems to be interpreted as *huaiyi*₁ without ambiguity.
- ▶ There is an unexplained lower acceptability for sentences with M-not-M operators denoting ‘can-NEG(-can)’ when the focus semantic value of the complement is a set with more than one member.
- ▶ The current discussion hasn’t involved situations of ‘*huaiyi*+constituent-wh complement’, which are actually important under the category of ‘*huaiyi*+interrogative complements’.

Remaining issues

- (33) a. 他们居然怀疑 这个 项目
they even *huaiyi*₁ this-CL program
的 可行性。
structural-PTCL feasibility.
'They even doubt the feasibility of this program.'
The chosen interpretation: *huaiyi*₁
- b. ?人们 甚至 开始 怀疑 奥运会 能 否
'People even started to suspect (*huaiyi*₂) that
[_F按时 召开].
the Olympics can happen on time.'
| [[能否_Q[奥运会[_F按时 召开]]]^F | > 1
The chosen interpretation: *huaiyi*₂

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